College of Computer and Information Sciences

# Computer Science Department

| CS1213-Design and Analysis of Algorithms | | 1st Semester 2025/2026 | | Quiz# 1 |
|---|---|---|---|---|
| Student Name | [redacted] | ID: | [redacted] | |
| CLOs: | 1.3 | Grade: | [ 4 ] / 10 | |

**(PLO#K1-CLO1.3) Question 1:** To be answered in **(20)** Minutes [ ] / 10 Marks

**Choose one correct answer for each statement and fill the following table: [1 point for each item]**

| Question | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| Answer | A | b | c | b | d | A | c | A |

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1. if $f(n) = 5.2\,n^3 + 20n + 1$ and $g(n) = 1.5\,n^3$, then:
   a. $f(n) = O(g(n))$
   b. $f(n) = \Omega(g(n))$
   c. $f(n) = \Theta(g(n))$
   d. All the above answers are correct.

2. What is the largest input size n that we can get it within a day if we have an algorithm with the running time $n$ and a computer with the performance 1000 operations per hour?
   a. $n = 1000$
   b. $n = 24000$
   c. $n = 3600$
   d. $n = 86400$

3. Given the functions $f(n) = 7n^5 + n$ and $g(n) = 5n^3$ which values of $c$ and $n_0$ can be used to prove that $f(n) = \Omega(g(n))$?
   a. $c = 5, n_0 = 2$
   b. $c = 100, n_0 = 2$
   c. $c = 10, n_0 = 1$
   d. All the above answers are correct.

$7n^5 + n \geq 5n^3$

4. If $f(n) = n^3 + n$, then $f(n)$ is:
   a. $O(n^3 + n)$
   b. $o(n^3 + n)$
   c. $\omega(n^3 + n)$
   d. All the above answers are incorrect.

5. If f(n) is not O(g(n)) then:
   a. $f(n) = O(g(n))$
   b. $f(n) = \Omega(g(n))$
   c. $f(n) = \Theta(g(n))$
   d. Cannot be determined.

6. Consider two different algorithms A and B. Both algorithms have been implemented using the same programming language. Using inputs with the same size, if the running time of A is 0.5 millisecond while the running time of B is 1 millisecond, then:
   a. A is faster than B
   b. B is faster than A
   c. A and B have the same speed
   d. Cannot be determined

7. If $f(n) = (2n^7 + n^4 + 10n^3)(3m^4 + 2m^3 + 1)$, then:
   a. $f(n) = O(n^{11})$
   b. $f(n) = O(n^7 m^4)$
   c. $f(n) = O(n^7)$
   d. All the above answers are incorrect.

8. If $f(n) = 3^n$ and $g(n) = \sqrt{n}^n$, then
   a. $f$ grows faster than $g$.
   b. $f$ grows slower than $g$.
   c. $f$ and $g$ have the same rate of growth.
   d. Cannot be determined.

**(PLO#K1-CLO1.3) Question 2:** To be answered in **(10) Minutes** [ ] / 2 Marks

**What is the time complexity of the following algorithm segments in the most precise Big O? [1 point for each item]**

| Segment A | Segment B |
|---|---|
| `j ← 1`<br>`for j<=n do`<br>`  k ←1`<br>`  for k<=j do`<br>`  print j`<br>`  k ←k+1`<br>`  endFor`<br>`  j ←j+1`<br>`endFor` | `i ←1`<br>`while i< n do`<br>`  if 0 < n`<br>`  j ←1`<br>`  for j<= n² do`<br>`  j ←i+1`<br>`  endFor`<br>`  else`<br>`  print n`<br>`  endIf`<br>`  i ←i*2`<br>`endWhile` |
| Answer: $O(n^2)$ | Answer: $O(n^2 \log n)$ |